Starter

Recall the disadvantages of a flat-file approach to databases:

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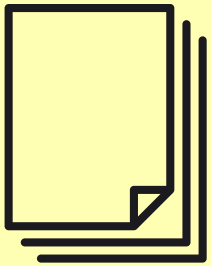

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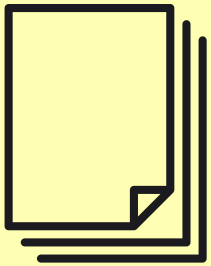

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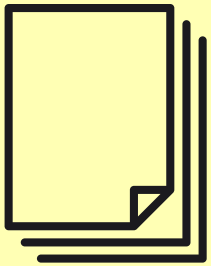

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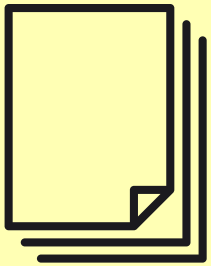

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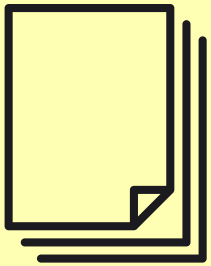

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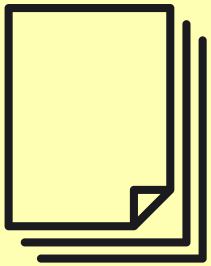

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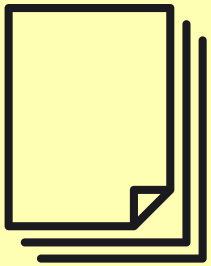

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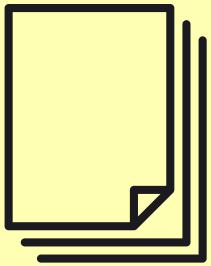

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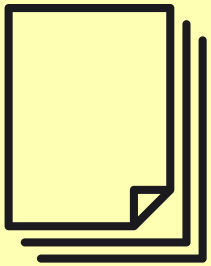

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5. Program-data dependence
   - File structure means that programs must know the structure of the entire system in order to interact at all. Changes to this structure will mean that programs must be changed. This is known as **unproductive maintenance**

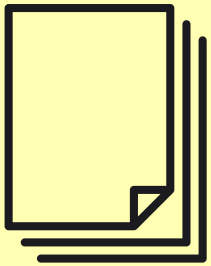
From 2018 Paper 2 Q7.3:

*Minimise data duplication // no unnecessary repeated data; A. reduce for minimise
R. eliminate
*Eliminate data redundancy; A. reduce/minimise for eliminate
Eliminate data inconsistency // improve consistency // avoid inconsistency problems;
Eliminate update anomalies; A. example in context A. updates only need to be made in one place
Eliminate insertion anomalies; A. example in context
Eliminate deletion anomalies; A. example in context
NE. easier to update/insert/delete without concrete example or good explanation
NE. fewer errors when updating/inserting/deleting without concrete example or good explanation
NE. saving space/memory NE. easier/faster to query

Database Design: Normalisation

Keywords: Flat file; Data inconsistency; data redundancy; program-data dependence; relational database; normalisation; primary key; foreign key; composite key; repeating group; 1NF; 2NF; 3NF;

Aim



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Relational Database

- A collection of tables linked by key fields

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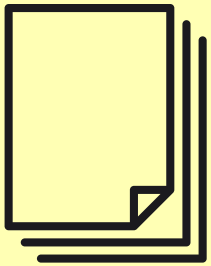

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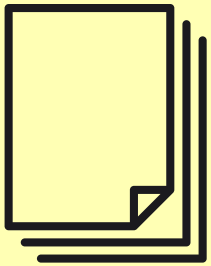
Normalisation

- The process of decomposing a flat-file database into the best* possible relational database.

* for storage size.

Example Normalisation



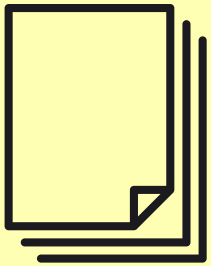
A publishing company want to create a database to keep data about their authors and each of the books that author has published. Each author has an agent, each published book has a category, and each book category has a specified royalty rate.

So we start with:

Author (Author number, Surname, Forename, Address, Phone number, Agent Number, Agent Name, Agent Address, Book Title, Book Category, Royalty Rate)

This is called **un-normalised form (UNF)**

First Normal Form (1NF)



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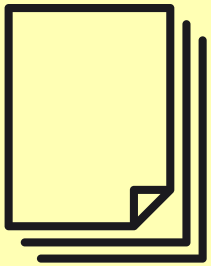

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This table now has two primary keys, known as a **composite key**.

Second Normal Form (2NF)

Any table with a **composite key** needs to be checked:

Do all of the non-key attributes depend on **every** part of the composite key?

If not, take them out and make a new entity.

Second Normal Form

1NF

Author (Author Number, Surname, Forename, Address, Phone number, Agent Number, Agent Name, Agent Address)

Book (Author Number*, Book Title, Book Category, Royalty Rate)

Second Normal Form

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2NF

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Publication (Author Number*, Book Title*)

Book (Book Title, Book Category, Royalty Rate)

Third Normal Form (3NF)

All non-key attributes must be checked:

Do they depend **only** on the primary key?

Each attribute must depend **on the key, the whole key and nothing but the key**

Third Normal Form

2NF

Author (Author Number, Surname, Forename, Address, Phone number, Agent Number, Agent Name, Agent Address)

Publication (Author Number*, Book Title*)

Book (Book Title, Book Category, Royalty Rate)

Third Normal Form

2NF

Author (Author Number, Surname, Forename, Address, Phone number, Agent Number, Agent Name, Agent Address)

Publication (Author Number*, Book Title*)

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Third Normal Form

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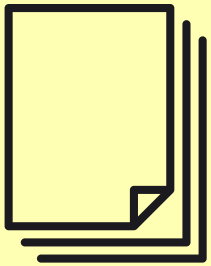

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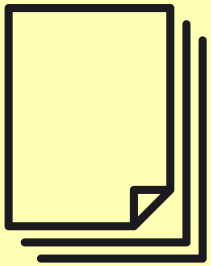

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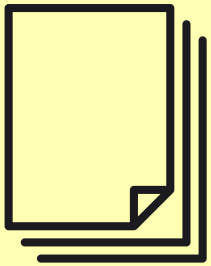

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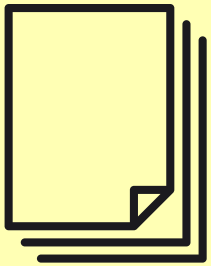

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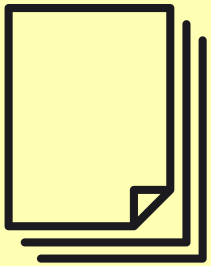

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L139 – Phoenix Circus School Table